Subset Sums in a Fourfold Sumset

Variable thresholds for persistent holes
and the audited bound $N \le 90$

Sum-of-Subset research project

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Abstract

We continue the independently audited bound $N \le 91$ by excluding cardinality 91. A variable threshold at each prefix improves the persistent-hole argument: at size 37 and sufficiently many holes, its defect floor rises from 144 to 193. A two-stage lattice comparison uses the one-budget bound first and reserves the stronger hole witnesses for the remaining range. This closes the inner lattice gate without a new defect-sensitive span lemma. The inverse theorem is used only when both supports are non-progressions; progression summands are handled by an exact gap formula, including a two-residue exception. We then reconstruct the complete necessary covering models at the new endpoint. Exact integer certificates and a standard-library verifier accompany the report. The stronger archived bound 84 is not asserted to have received a complete independent replay.

1 Result and the inherited reduction

For a finite real set A, $\mathrm{FS}(A)$ denotes all subset sums, including zero, and $4A = A + A + A + A$ allows repetitions. Let N be the largest cardinality for which $\mathrm{FS}(A) \subseteq 4A$.

Theorem 1.1. *Every feasible set has at most 90 elements. Hence* $71 \le N \le 90$.

The lower construction remains $([-31, 31] \cap \mathbb{Z}) \cup \{\pm 93, \pm 155, \pm 217, \pm 713\}$. The previous audited bound is 91 [1]. Thus it suffices to exclude size 91, and such a set must contain zero: otherwise adjoining zero preserves feasibility and gives 92 elements. We use the corrected integralization, covering, and peeling lemmas in [2]. No inheritance of feasibility by arbitrary subsets is assumed.

Write $T_r = r(r+1)/2$. For a positive r-set X, let

$$\mu(X) = |\mathrm{FS}(X)| - 1 - T_r, \qquad \rho(X) = \mu(X) + 2 - 2\delta_0(X),$$

where δ_0 counts x for which $2x$ is absent from the restricted twofold sumset of $X \cup \{0\}$. We inherit $\rho \ge -8$ universally, $\rho \ge 0$ for $r \ge 14$, and $\rho \le \mu$ because the maximum is defective. Non-progression support has $\mu \ge r - 3$; its defect is at least $r - 14$ for $r \ge 6$ and at least $r - 5$ for $r \ge 17$ [3]. Let $f(r)$ denote the latter available floor.

Remove four original extremes on each side to obtain D, and perform the two four-extreme peels $D \supset D_1 \supset E$, where $E = P \cup \{0\} \cup (-Q)$. At the present endpoint,

$$d = 83, \quad e = 75, \quad a + b = m = 74, \quad I = (a - 37)^2, \quad \Lambda = 1 + T_a + T_b = 1407 + I, \quad \Omega = 516 - I.$$

The exact inherited budget is

$$\rho + 2\vartheta + c + \mathcal{T} = \Omega, \qquad c = 2|2D| + 18d + 85 - |\mathrm{FS}(D)|, \qquad \mathcal{T} = \kappa + \ell_1 + \ell_2 + g, \tag{1}$$

where $\rho = \rho(P) + \rho(Q)$, $\mu = \mu(P) + \mu(Q)$, and $\kappa = |\mathrm{FS}(P) - \mathrm{FS}(Q)| - |\mathrm{FS}(P)| - |\mathrm{FS}(Q)| + 1$. All displayed slacks other than ρ are nonnegative.

The deficient-sign check is $1 + T_{75} = 2851 > 24 \cdot 91 + 77 = 2261$. The paired bootstrap gives $q^2 \le 2648$ with even q, so $q \le 50$ and $a, b \ge 12$. At a side of size 12 or 13, respectively, $\Omega = -109$ or -60, below the general lower bound -8 for the combined defect. These sizes are impossible. Both sides are now at least 14; nonnegative defect excludes size 14, where $\Omega = -13$. Thus $15 \le a \le 59$, giving 45 oriented lattice cases.

2 Variable thresholds for persistent holes

We first prove the new general estimate. It uses only inherited positive-side lemmas and the full-prefix central interval from [1], itself a direct application of Lev's Theorem 1 [7]. The latter states that an i-set in $[1,l]_{\mathbb{Z}}$ with $l \le 3i/2-2$ fills all subset sums between $2l-2i+1$ and its complement about half the total. Consequently, if Y has i elements, $\max Y \le i+\beta$, and $i \ge 2\beta+4$, then

$$\mathrm{FS}(Y) = [0,S]_{\mathbb{Z}} \setminus (H \cup (S-H)), \quad H \subseteq [1,2\beta], \quad |H| \le \beta, \quad S = \sum Y. \tag{2}$$

Indeed apply Lev with $l = i+\delta$, where $0 \le \delta \le \beta$; each missing lower sum is a missing coefficient in $[1,2\delta]$.

Theorem 2.1 (A schedule of small-increment thresholds)**.** *Let X be a positive integer r-set with hull-hole count $z \le h_X < |\mathrm{FS}(X)|-1$. Choose integers $18 \le L \le r$, $B \ge 0$, and $\beta_i \ge 2$ for $L \le i \le r$. Suppose*

$$2\beta_i < z, \qquad 2\beta_i + 4 \le i, \tag{3}$$

$$B \le T_{L-1} - r - 1, \qquad B \le T_i + i - 6\beta_i - r - 7 \quad (L \le i \le r), \tag{4}$$

$$B \le L - 6 + \sum_{i=L}^{r} (\beta_i - 1). \tag{5}$$

Then $\rho(X) \ge B$.

Proof. Assume $\rho_r < B$. The strict upper hole condition forces X to be primitive; a gcd at least two would create at least $|\mathrm{FS}(X)|-1$ holes. Its support is non-progression because $h_X > 0$. For increasing prefixes X_i, put $y_i = |\mathrm{FS}(X_i)| - |\mathrm{FS}(X_{i-1})| - i \ge 0$. We have $\rho_i \ge \rho_{i-1} + y_i - 2$, and $y_i = 0$ at $i \ge 6$ forces progression support.

If the last progression prefix has size $t \ge L-1$, its next translate is disjoint: an on-lattice overlapping or abutting translate would preserve progression support. Thus $y_{t+1} \ge T_t - t$. All later prefixes are non-progressions, so their increments are at least one. Starting with $\rho_t \ge 0$ gives $\rho_r \ge T_t - r - 1 \ge T_{L-1} - r - 1 \ge B$, a contradiction. Every prefix from $L-1$ onward is therefore non-progression, and $\rho_{L-1} \ge L-6$. If $y_i > \beta_i$ for every $i \ge L$, the recurrence and (5) already imply $\rho_r \ge B$.

Choose an $i \ge L$ with $1 \le y_i \le \beta_i$. The inherited small-increment inverse applies since $i \ge 2\beta_i + 4 \ge \beta_i + 5$. It writes $X_i = tY$ with Y primitive, $|Y| = i$, and $\max Y \le i + \beta_i$. Here t is an integer. If $t > 1$, final primitivity forces a later first element off $t\mathbb{Z}$. Its translate is disjoint, and the preceding calculation gives $\rho_r \ge T_i - r - 1 \ge B$. Hence $t = 1$ and (2) applies with the fixed threshold β_i chosen at this prefix.

Propagate its endpoint-hole description through later prefixes. Suppose it holds at size $v \ge i$, with total S, and the next element is x. If $x \ge S - 4\beta_i$, the translate adds at least $S - 6\beta_i$ new points above S: for $x \le S$ the last x positions lose at most $2\beta_i$ holes, while for $x > S$ the whole translate is new. Using $\rho_v \ge v - 5$ and later increments at least one yields

$$\rho_r \ge S + v - 6\beta_i - r - 7 \ge T_i + i - 6\beta_i - r - 7 \ge B.$$

Thus $x < S - 4\beta_i$. Also $x > 2\beta_i$ by increasing order. The central intervals overlap or abut, the lower holes remain H, and complement symmetry supplies the upper holes. Induction gives $h_X \le 2\beta_i < z$, the final contradiction. □

The threshold changes only in the search for the useful prefix. Once that prefix is selected, its own β_i remains fixed during hole propagation. This distinction is essential.

For a proposed B and L, the largest permitted local thresholds are

$$\beta_i(B) = \min\left\{ \left\lfloor \frac{i-4}{2} \right\rfloor, \left\lfloor \frac{z-1}{2} \right\rfloor, \left\lfloor \frac{T_i + i - r - 7 - B}{6} \right\rfloor \right\}. \tag{6}$$

The checker requires every threshold to be at least two and verifies (3)–(5) exactly. Feasibility is monotone as B decreases, so the optional witness search uses integer bisection. The theorem needs any valid schedule, not optimality of that search. The previous constant-threshold bound is retained if stronger.

At $(r, z) = (37, 215)$, take $L = 22$, $B = 193$, and

$$(\beta_{22}, \dots, \beta_{37}) = (6, 9, 10, 10, 11, 11, 12, 12, 13, 13, 14, 14, 15, 15, 16, 16).$$

The AP escape bound is exactly 193, the accumulation is 197, and every local jump bound is at least 193. The previous bound was 144. The same schedule already works for $z \geq 33$. At $z = 15, 19$, respectively, the new floors are 126 and 152, compared with 123 and 144. The helper `verification/nonap_hole_rho_schedule.py` records all schedules, checks 3,311 parameter queries (2,480 strictly improved), and retains the previous finite central-interval audit. A separate deterministic check computes actual supports and diagonal defects for 2,432 integer sets, including 1,102 with the required strict hole condition. These finite checks supplement the general proof above.

3 A complete progression-support repair

The raw sufficient inverse margin is now $J = T_{\min(a,b)} - 2 - \Omega$, with minimum -49. We use Stanchescu's Corollary 2 [8] only under the sufficient strict condition $\kappa < \min(|\operatorname{FS}(P)|, |\operatorname{FS}(Q)|) - 3$. If both supports are non-progressions, put $r = \min(a, b)$ and $t = 74 - r$. The smaller-support margin is at least

$$J + (r - 3) + f(r) + f(t) \geq 36,$$

and the larger baseline margin $T_t - 2 - \Omega$ is at least 185. This gives a common core lattice, side holes $h_P, h_Q \leq \kappa$, and strict side-hole bounds $h_X < |\operatorname{FS}(X)| - 1$. Both positive sets are integral on this lattice.

If a support is a progression, we instead use an elementary exact formula. For $W = \{w_1 < \cdots < w_v\} \subset \mathbb{Z}$ and $J_u = [0, u-1]_{\mathbb{Z}}$,

$$|W + J_u| = v + u - 1 + \sum_{i=1}^{v-1} \min(w_{i+1} - w_i - 1, u - 1). \tag{7}$$

Add the translated intervals in increasing order to prove this. For a progression of step t, apply the formula in each occupied residue class. If there are s classes, the excess is at least $(s-1)(u-1)$.

We also need a subset-sum observation. For positive $x_1 < \cdots < x_b$, the largest consecutive gap in the support is $\max_i(x_i - \sum_{j<i} x_j)$. Indeed adding $[0, L-1]$ fills one interval precisely when $x_i \leq L + \sum_{j<i} x_j$ for every i, by induction; a failed test leaves a gap that later positive elements cannot fill. If exactly one support gap has length at least u, a prefix gap attaining the maximum and its complement reflection must be the same gap. Their endpoints force $i = b$. The last coefficient dominates the previous total, and therefore

$$\mu(X) \geq 2(1 + T_{b-1}) - 1 - T_b = T_{b-2}. \tag{8}$$

Call the progression $U = t[0, u-1]$, and the other support V. If its side has at least 37 coefficients, then $u - 1 \geq T_{37} = 703 > \Omega$. Formula (7) forces V into one residue class and forbids any gap of length at least u. On that lattice $h_U = 0$, $h_V = \kappa$, and $z = 0$.

It remains that the progression side has $15 \leq a \leq 37$ coefficients, and the other side has $b = 74 - a \geq 37$. Exact quadratic arithmetic gives

$$\Omega < 2T_a, \qquad \Omega - T_a \leq 47, \qquad T_a + 1 > 2b. \tag{9}$$

There are at most two residue classes in V. In one class, there can be at most one gap of length at least u. Such a gap would give (8) and $\rho(Q) \geq T_{b-2} + 2 - 2b \geq 558 > 516$, impossible. So this case also gives $h_U = 0, h_V = \kappa, z = 0$.

In two classes, one off-lattice coefficient would give two disjoint copies of a $(b-1)$-element subset-sum support and the same contradiction. There are at least two off-lattice coefficients. Their common nonzero residue must have order two, since their sum is also represented. Normalize the half-step so that $U = 2[0, u-1]$ and Q has at least two odd coefficients. Let w be its least odd coefficient, and let h_0, h_1 be the two fiber hull-hole counts, measured with step two. No fiber gap reaches u in its own coordinates: the baseline excess $u-1$ plus that gap would contradict (9). Formula (7) and complement symmetry give

$$\kappa = u - 1 + h_0 + h_1, \quad h_U = u - 1, \quad h_Q = \kappa - u + w, \quad z = w - 1. \tag{10}$$

The $w-1$ term counts the integer positions outside the two fiber hulls. These identities hold whether the total of Q is even or odd.

We first show $w \le u$. If $w \ge u+1 > 2b$ and $s \ge 2$ coefficients are odd, distinctness gives

$$\sum Q \ge (b-s)(b-s+1) + sw + s(s-1).$$

This quadratic is increasing for $s \ge 2$. Using (10), $|V| \ge b^2 - 3b + 2u + 6 - \kappa$. But $\mu_Q + \kappa \le \Omega + 2b - 2$, by (1) and $\rho_Q \ge \mu_Q + 2 - 2b$, gives $|V| \le T_b + \Omega + 2b - 1 - \kappa$. The lower minus upper margin is at least $(b^2 - 11b)/2 + 2(T_a + 1) + 7 - \Omega > 0$ in every case. Thus both side holes in (10) are at most κ.

In fact $w \le 13$. For odd $w \ge 15$, minimizing the same mass quadratic over real s gives

$$\sum Q - w - T_b \ge 7b - 289/8.$$

On the other hand $\mu_Q + \kappa = \sum Q + u - w - T_b$ and (9) bound this expression above by $2b + 44$, a contradiction for $b \ge 37$. Therefore $z \le 12$.

The progression side satisfies $R_P \le 2(a + \mu_P)$ by subtracting $2T_{a-1}$ from its total. The other side satisfies $R_Q \le b + \mu_Q + \kappa - T_a - \mu_P + 12$ by subtracting T_{b-1}. Insert these bounds into the exact pre-lattice comparison, and use $\mu \le \rho + 2m - 4$ and $\rho + \kappa + \ell_2 \le \Omega$:

$$|\operatorname{FS}(D_1)| - 2|\operatorname{FS}(E)| \le 2543 + 2a - 4T_a - 2T_b \le -1157.$$

The outer comparison has margin at least 569. Off-lattice points would give disjoint support translates, so D_1, D lie on the half-step lattice. Only now may we use the sharp peeling bound $\mathcal{T} \ge \kappa + 156 - 3z \ge T_a + 120$, contradicting $\Omega - T_a \le 47$. Its margin is at least 73. This excludes the two-residue exception without a circular use of post-lattice peeling.

The progression cases left have $z = 0$. The general span lemma $\max X \le |X| + \rho(X) + h_X + 2$ for integer sets of size at least 15 requires no strict side-hole condition [1]. It gives inner comparison $C + \Omega < 0$ with $C = -945 - I$, and the same outer comparison. After lattice propagation, $\kappa \le \Omega - 156$, which gives $T_{\min(a,b)} - \kappa \ge 109 > 0$. Strict side-hole bounds are now available in these cases as well. This progression repair rederives the analytic size-91 argument in the archive [6]; none of its endpoint certificate files is used here.

4 A two-stage lattice comparison

For the non-progression case the core is already normalized, so the new hole theorem is available before the peeled layers are integral. Write $s = |\operatorname{FS}(E)|$ and $z = h_P + h_Q - \kappa$, so that $0 \le z \le \min(h_P, h_Q)$ and both strict side-hole bounds hold. The inherited inner comparisons are

$$|\operatorname{FS}(D_1)| - 2s \le \min\{C + \Omega + \rho - \mu + 2z,\ C + 3\Omega - \rho - \mu\}, \quad C = -945 - I. \tag{11}$$

Since $\rho \le \mu$, the first is $-429 - 2I + 2z < 0$ whenever $z \le 214 + I$. In the remaining range $z \ge 215 + I$, first use the non-progression floors. Any unresolved orientation has $a, b \ge 26$. For each side choose the best inherited effective-hole witness $\mu(X) \ge u_r = r + C_h + 1$ with $2\beta < 215 + I$. Its two strict hypotheses are checked again by the verifier. Combine this with

$$\rho(X) \ge \max\{F(r, u_r), \mathcal{R}(r, 215 + I)\},$$

where F is the exact prefix inverse [4] and $\mathcal{R}$ is the maximum of the old and new persistent-hole floors. All 45 oriented comparisons are strictly negative: the first-stage maximum is -1, and the remaining second-stage maximum is -13. At balance the previous constant-threshold bound left the comparison at $+3$. Now $u_{37} = 156$ and $\mathcal{R}(37, 215) = 193$, giving $-945 + 3 \cdot 516 - 2(156 + 193) = -95$.

The outer margin is $\Lambda - (4d - 10 + \Omega) = 569 + 2I > 0$. Thus both peeled layers lie on the core lattice. Hole propagation and sharp four-extreme peeling [5] now give

$$w \le 4z, \quad \mathcal{T} \ge \kappa + 156 - 3z, \qquad \mathcal{T} \ge \max(52, z, 156 - 2z). \tag{12}$$

No state with a side of size 15 or 16 survives these budgets, so all later inverse and hole-helper calls are within the range $r \ge 17$.

5 Covering models at the new endpoint

We preserve the actual-set-to-model construction from [1]: the same eighteen ordinary units, first actual contributor orientation, split-pair choices, integer row-lattice witnesses, loss variables, subgroup overlap inequalities, six core cutoff positions, and 85 trimmed sumset rows. The possible empty-unit count is still at most four. All 10,880 four-empty modes are retained, including the 30 modes with only partial integer gap information. No rational division is used to infer an integer coordinate.

The four low-hole families are

Family	z	$\rho_P + \rho_Q \ge$	$\mu(X) \ge$	$c + I \le$
Z	0	0	0	360
P	1–2	64	$r - 3$	300
Q	3–4	76	$r + 6$	292
R	5–14	97	$2r - 5$	291

The last three caps equal $360 - \rho_{\text{floor}} + 2z_{\text{endpoint}}$. The effective-hole rows and exact prefix inverse certify the Q and R defect floors for $17 \le r \le 57$. Each positive family transports z to its own endpoint Z and adds $Z - z$ to both mass coordinates. Every source row is checked for the required nonpositive change; the auxiliary masses need not describe another actual set. For empty count j, the low parameter interval is exactly the set of integers a satisfying $(a - 37)^2 \le C_{\text{family}} - 83j$. These envelopes contain 369,640 integer low roots, including any already impossible scalar states; enlarging the necessary model domain is harmless.

For $z \ge 15$ the scalar enumeration leaves 1,320 states; $c \le 140 < 2d$, so only $j = 0, 1$ occur. The all-active comparison leaves 196 states at $j = 0$ and the empty-unit budget leaves 66 at $j = 1$. They form 42 groups with fixed side sizes and support/defect floors. For comparison, the same post-lattice scalar relaxation with only the previous constant-threshold floor has 3,596 states, cap 176, and 56 two-empty states; the variable thresholds reduce these to 1,320, 140, and zero. This comparison alone is not an endpoint exclusion. The same real variable z is retained across each group's full interval. With $E_* = \Omega - \rho_p - \rho_q$ and $\delta = c - 83j$, use all four caps

$$\delta \le c_{\max} - 83j, \quad \delta + z \le E_* - 83j, \quad \delta \le E_* - 52 - 83j, \quad \delta - 2z \le E_* - 156 - 83j.$$

The verifier checks 109,514 coarse-source and 211,432 nested-source row implications, including every new full-translate row. It does not combine unrelated maxima of c and z, nor choose the reflected all-active orientation independently of the actual contributor.

For clarity, all dimension-dependent replacements in the inherited covering model are listed below. Put $K = |2D|$, let R_s, C_s be the largest and second largest core magnitudes, and let M_s be the largest magnitude in D.

Quantity	New value
Ordinary unit capacity d; core size e	83; 75
Global quantile pair; core quantile pair	83/82; 79/78
Ordinary quantile pair	opposite size +5 / own size +4
Basic leading threshold $2d-1$	165
Nested leading threshold $2d-9$	157
Second-core leading threshold $2d-11$	155
Core loss threshold $d-4$	79
Span budget $R_p+R_q+\delta-4z$	$438-I-83j$
Cardinality budget $S_p+S_q+\delta-2K-4z$	$1578-83j$

Each threshold is applied to its own proved residual cap and charged leading side. The trimmed increment remains $83s-T_{s-1}$ for removing s extreme elements. The coarse bound remains $K \le M_p+M_q+R_p+R_q+21$. For high groups, $140<155$ licenses the deeper leading quantiles; after one empty unit, $140-83=57<79$ licenses core quantiles. The actual selected cap, not these global maxima, determines each row.

The coarse model is tried first. When needed, the nested model and the inherited exhaustive disjunctions are used: density, ordinary versus outside-center quantiles, core-loss alternatives, six inner positions, and both-or-sole contributors for a split pair. Every binary child must be present. No solver-selected cut is accepted unless reconstructed from one of these proved alternatives.

5.1 Using the location of an empty own translate

There is one additional low-hole refinement. At $z=0$, the target is the full interval $[1,S_p]_{\mathbb{Z}}$ on the positive side. Suppose an integer own center $u=c_p+p_{i+1}$ is empty, but a lower own center $v=c_p+p_{h+1}$, $h>i$, contributes. Then $u>v$ and $u>S_p$. Indeed $0<u\le S_p$ contributes through $u+0$. If $u\le 0$, choose $d'\in D$ with $1\le v+d'\le S_p$. Then $1\le u+d'\le d'\le M_p\le S_p$, also a contribution. Thus the new necessary row is

$$-u+S_p\le -1.$$

Reflection gives the negative-side row. We activate it only for zero-hole groups with a complete integer-center basis, an explicitly missing own single, and a retained lower own single.

When $i=0$, the four original fixed labels $C^+_{1j}=c_p+p_1+p_j$ $(1\le j\le 4)$ all exceed S_p. They therefore contribute four additional fixed losses. The original fixed cover has 110 labels and 26 disjoint groups, each supplying a nonpositive label. These four high labels are positive, so they cannot be any of those nonpositive deletions; their losses are additional. The reflected four labels are separate labels. Counting labels rather than distinct values only enlarges the covering upper bound and is safe.

Let f be four times the number of sides meeting this largest-empty condition. Then both the ordinary loss sum and the combined subgroup loss sum are at most $\delta-f$. The same correction is made in the sole-contributor branch. It is one loss charged consistently in alternative necessary inequalities, not a sum of repeated gains. These deductions also rederive the applicable fixed-label argument in [6]. The verifier reconstructs the original 110-label/26-group cover and checks 10,430 small configurations, including 8,739 high-center tests and 22,952 high fixed-label tests. The arbitrary-size proof is the interval argument just given.

5.2 Full translates, a common deficit, and clipping

The nested model also uses all of D in selected overlap comparisons, rederiving the full-translate argument in [6]. On the positive side let the leading component be c_p+2D, and select own components c_p+p_i+D and cross components c_p-q_j+D. All these centers are integral in the modes where these rows are used. If k own and l cross components are selected, their total nominal capacity is $C=K+(k+l)d$. Let p_*,q_* be their largest selected outer magnitudes when present.

The union lies in $[L, U]_{\mathbb{Z}}$, with

$$L = c_p - \begin{cases} q_* + M_q, & l > 0, \\ 2M_q, & l = 0, \end{cases} \qquad U = c_p + \begin{cases} p_* + M_p, & k > 0, \\ 2M_p, & k = 0. \end{cases}$$

For the actual target $F_p = \mathrm{FS}(D) \cap (0, \infty)$ and subgroup union W, put $G = C - |W \cap F_p|$. Every member loss λ_v satisfies

$$G \geq C - (U - L + 1) + \lambda_v, \qquad G \geq \sum_v \lambda_v.$$

Indeed $C - |W \cap F_p| = (C - |W|) + |W \setminus F_p|$, and the outside union contains each member's outside set. Nonemptiness of a selected ordinary component further gives the clipped bounds

$$G \geq C - U, \qquad G \geq C - S_p + L - 1, \qquad G \geq C - S_p.$$

These are bounds for the actual target deficit; no selected member loss is added to a clipped bound.

For each side, the larger family consists of the leading component, all retained own components, and retained unpaired cross components with labels 6 and 7 (outer indices 3 and 4). Its variable Γ_p is the actual capacity deficit of the whole family. For any selected subgroup, Γ_p dominates its deficit plus the individual losses of members outside that subgroup. Taking suffixes of the own and cross lists gives all implemented mixed rows. The two larger families use disjoint component labels, so

$$\Gamma_p + \Gamma_q + \sum_{\text{unmatched } v} \lambda_v \leq \delta - f.$$

The earlier block inequalities remain separate necessary inequalities; their gains are not added to this new allocation. All-subset own-only rows analogously strengthen the earlier own-block variables. The 49-set sanity audit checks 795 subgroup inequalities. Positive-hole mass transport preserves these rows because each mass coefficient is zero or negative.

5.3 Exact fixed-label indicators in the four-empty exception

Only the zero-hole four-empty mode with missing units `pair1,pair2,p0,q0` enables the fixed-score extension, in either structural branch. It already charges $f = 8$. For each of the other 102 original fixed labels v, introduce an auxiliary $H_v \in [0, 1]$, realized as the indicator that the score exceeds its own target mass. For all 110 labels introduce $N_v \in [0, 1]$, realized as the indicator that the score is nonpositive. Each of the 26 disjoint negative groups requires the sum of its N_v to be at least one. Thus

$$F_{\text{extra}} = \sum_{v \text{ not already charged}} H_v + \sum_v N_v - 26 \geq 0$$

is an additional actual fixed loss. The eight already charged labels are omitted from the high sum. High and nonpositive events are disjoint, so this loss may be added to each complete deficit allocation.

All score forms are integer combinations of the eight proved integral outer coordinates, and both masses are integral in these zero-hole groups. For a score f_v with target mass S, the exhaustive splits are

$$(f_v \leq S,\ H_v = 0) \quad \text{or} \quad (f_v \geq S + 1,\ H_v = 1),$$
$$(f_v \leq 0,\ N_v = 1) \quad \text{or} \quad (f_v \geq 1,\ N_v = 0).$$

Their actual indicator realizations justify the necessary relaxations and both branches. Universally positive score forms, recognized by nonnegative coefficients in positive outer-gap coordinates, have $N_v = 0$ directly. The eight charged high labels also have $N_v = 0$. The helper `verification/fixed_scores.py` regenerates every form from the covering definitions, transforms each of the 26 group sums to gap coordinates, checks its nonpositive coefficients, and checks disjointness. Stored coefficient vectors alone are not trusted.

6 Exact certification and reproduction

The certificate bundle contains 995 distinct nonnegative integer multiplier vectors. It covers 27,802 structural groups, 371,986 integer roots, and 372,137 final leaf systems. Of the 27,802 groups, 27,250 close in the coarse model and 552 use the nested model. There are 27,836 symbolic interval checks, 80,170 integer extremum evaluations, and 25 tree roots with 151 binary nodes and 176 tree leaves. The maximum tree depth is 29. The splits comprise 18 core-loss, 13 inner-position, 13 high-score, and 107 nonpositive-score alternatives. The minimum normalized strict contradiction gap is 3/151. For every leaf with variables $x \geq 0$ and necessary rows $Ax \leq b$, the checker verifies $w \geq 0$, $wA \geq 0$, and $wb < 0$ using integers and exact rational arithmetic. The generator may use SciPy to discover w; no floating-point result is accepted as a proof.

The checker reconstructs the whole root universe, each allowed disjunction, and all parameter intervals. Quadratic right sides are maximized at endpoints and the nearest integers to any interior vertex. The exhaustive option additionally instantiates every integer parameter and compares every row with its symbolic formula. The high-hole rows cover real intervals and hence all actual integer hole counts. Negative controls reject 43 deliberately malformed proof inputs, including missing roots, omitted children, invalid integer division, nonstrict schedule thresholds, and incomplete schedules.

From the repository root:

```
python -I -S -B verification/upper90.py --output .work/upper90.json
python -I -S -B verification/upper90.py --exhaustive \
  --output .work/upper90-exhaustive.json
python experiments/generate_upper90_certificates.py \
  --output .work/upper90-regenerated.json
```

Only the optional generator needs NumPy and SciPy. The Windows entry point is `verify_upper90_windows.bat`. Complete run receipts, source hashes, scalar witnesses, and compressed full output are under `results/2026-10-18/upper90/`. Ordinary and optimized-Python runs, full integer expansion, independent certificate regeneration, and the unchanged upper-91 baseline are recorded separately.

The result is a continuation of the audited chain, not a claim that the archived bound 84 is false or has been completely rechecked. The arbitrary-set mathematical lemmas remain part of the proof; the program is an exact finite certificate checker, not a proof-assistant formalization of the entire argument.

References

[1] Sum-of-Subset project, *Anchored hole bounds and integer center bases, and the audited bound $N \leq 91$*, `reports/upper91_from92_20261017.tex`.

[2] Sum-of-Subset project, *From the audited bound 115 to 103*, `reports/upper103_from115_20261014.tex`.

[3] Sum-of-Subset project, *Sharp non-progression defects and the audited bound 94*, `reports/upper94_from95_20261016.tex`.

[4] Sum-of-Subset project, *An integer prefix inverse and the audited bound 93*, `reports/upper93_from94_20261016.tex`.

[5] Sum-of-Subset project, *Sharp peeling and joint covering losses*, `reports/upper95_from96_20261016.tex`.

[6] Sum-of-Subset project, *Complete report for the archived upper bound 86*, section "A common core lattice at size 91", `reports/upper86_complete_20261005.tex`.

[7] V. F. Lev, *On consecutive subset sums*, Discrete Mathematics 187 (1998), 151–160, Theorem 1. doi:10.1016/S0012-365X(98)80006-X.

[8] Y. Stanchescu, *On addition of two distinct sets of integers*, Acta Arithmetica 75 (1996), 191–194, Corollary 2. doi:10.4064/aa-75-2-191-194.